Luis Ángel Quiroz López

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Professional Summary

Physicist and data scientist with a strong foundation in quantitative modelling, statistical analysis and production-oriented Python workflows. Works across the modern data stack (Python, pandas, SQL, Spark, PyTorch, TensorFlow, Azure ML).Contributed directly to LLM training pipelines through RLHF and human-in-the-loop annotation, designing rubrics and metrics to evaluate model outputs, and independently built a small language model in PyTorch end to end, from a from-scratch transformer and tokenizer through SFT, DPO and RL-based reasoning fine-tuning.Built end-to-end retail analytics projects covering SQL extraction and cleaning, star-schema modelling, and executive interactive HTML/JS KPI dashboards on delivery compliance, cycle times and store-level bottlenecks, published as a documented repository.Applied the same rigor to financial markets, cross-validating an EMA/MACD signal toolkit between Python and Excel as part of ongoing coursework in stock market analysis and technical indicators.
Currently completing an M.S. in Particle Astrophysics, applying statistical and computational methods to large astronomical catalogues and cosmological models. Fluent in English and Spanish, conversational in German, Italian and Portuguese.

Core Skills

Programming languages & data: Python (Pandas, NumPy), SQL, R, Spark, Bash, HTML/CSS and JavaScript.
ML/AI: PyTorch, TensorFlow, Keras, scikit-learn, LLM evaluation (rubric design, red-teaming, output scoring), RLHF, human-in-the-loop annotation, NLP, generative AI
Cloud & tooling: Azure ML, Snowflake, Flask, FastAPI, Git, Claude Code, Linux (Ubuntu)
Analysis & reporting: Chart.js, Excel, Mathematica, MATLAB, LaTeX, Power BI
Additional Software: Microsoft Office Suite, Canva, Photoshop, Audacity, CapCut, Davinci Resolve, Premiere Pro.

Languages: Spanish (native), English (C2 pursuing an English Proficiency Cambridge certification), German (B1/intermediate), Italian (B2/intermediate), Portuguese (B1/intermediate).

Selected Projects

Retail Logistics Analytics – *Python, Pandas, SQL, Chart.js, GitHub Pages* – : End-to-end delivery-compliance case study on a synthetic retail dataset of 62,420 shipment records across 61 stores, cleaned to 59,615 KPI-valid rows with every exclusion documented. Modelled the data as a star schema, benchmarked stores against same-zone peers rather than the global mean, and applied Wilson intervals to avoid ranking noise. Quantified value at risk by segment and deployed an interactive dashboard on GitHub Pages.
MACD Signal Toolkit – *Python, Pandas, yfinance, Excel, pytest* –: Built an EMA/MACD calculation engine for a BMV-listed equity, implemented in parallel as a Python CLI and an Excel workbook of 3,815 linked formulas. Cross-validated both implementations to within 5×10^{-13} across EMA, SMA, MACD, signal line, histogram and RSI, confirming matching crossover detection between the two. Covered with 15 automated tests and a written report explaining the EMA warm-up bias and the minimum sample size a signal line requires.
Small LLM from Scratch – *Python, PyTorch* –: Implemented an autodiff engine from first principles, then a decoder-only transformer component by component (masked multi-head attention with KV caching, RMSNorm, gated MLP), a byte-pair-encoding tokenizer trained from scratch, and the Adam optimizer, and used the full stack to train and sample from a small language model on TinyStories. Extended the pipeline through post-training, fine-tuning with SFT and DPO on chat-formatted data, and training a reasoning model with SFT followed by reinforcement learning on GSM8K math problems.
Galactic Structure of the Milky Way – *Python, TOPCAT* –: Cross-matched Gaia DR3 with APOGEE-2 DR17 and reduced 682,284 sources to 281,612 through documented quality thresholds on SNR, RUWE and parallax error. Classified kinematic components using Toomre diagrams and velocity ellipsoids, detected moving groups with DBSCAN, and derived the rotation curve between 6 and 12 kpc.
Neural network for Galaxy Morphology Classification – *Python, TensorFlow, Keras* –: Convolutional neural network trained on approximately 5,600 galaxy images with imbalanced classes (3,530 late-type against 2,080 early-type), with the imbalance addressed explicitly in both the training procedure and the evaluation metrics.
Stellar Population Statistics – *R* –: Maximum-likelihood fits comparing exponential and gamma models with AIC-based model selection over 240 observations, Box-Cox and Ordered Quantile normalisation, and hypothesis testing across 15 population pairs.
Personal Portfolio and Technical Log – *HTML, CSS, JavaScript* –: Hand-coded portfolio site with a draggable multi-window interface and embedded project dashboards. lqcorner.neocities.org

Experience

- **AI Trainer - Software Validation & NLP at DataAnnotation (2026)**:
  - Delivered high-quality annotations for Natural Language Processing (NLP) tasks across projects, supporting model evaluation and fine-tuning
  - Applied Reinforcement Learning Through Human Feedback (RLHF) with linguistic and contextual feedback to refine model behaviour, improving response quality in real-time evaluations.
- **Language Specialist/Freelance AI trainer at Meridial Marketplace (2026)**:
  - Contributed to human-in-the-loop pipelines improving LLM performance on multilingual tasks.
  - Reviewed and corrected model outputs, providing structured feedback used to guide training.
- **Field Data Verification Freelance Specialist (2026)**: Executed multimedia data collection and verification for commercial venues through on-site visits and digital documentation, ensuring dataset accuracy and completeness.
- **Web Developer - Freelance work at Ashland Chemical de México (2025)**: Built an interactive product-presentation website for private-school outreach, translating client requirements into a deployed UI.

Research & academic achievements

- **Master's Thesis Project *Effective Equation of State and Gravitational Particle Production in Inflationary Models* (2026)**: Supervised by Dr. Marcos A. García García, Institute of Physics, UNAM. Modelled inflaton and spectator-field dynamics during the reheating transition using numerical simulation and statistical inference in Python.
- **Research Assistant (2024)**: Literature review on cosmology-related topics under the supervision of Dr. Gabriella Picchinelli Bocchi and Dr. Ángel Sánchez Cecilio. Delivered weekly presentations on theoretical aspects of cosmological perturbations.
- **First place award at ENP Interpreparatory Contest (2017-2019)**:
  - Won the regional Astronomy category by presenting the *Knowledge and Skill* exam.
  - Won the regional *Italian II (Il mio primo racconto in podcast)* category by writing and producing a podcast.
  - Won local Physics category by writing the essay *Natural and Artificial Satellites*.
- **III Colloquium of the Young People Towards Research in Social Sciences, Humanities, and Arts Program (2018)**: Directed social science research and authored the research essay *Why Study Philosophy Today?*
- **Participation at Domino TIC-ENP Student Fair (2018)**: Presented the project *Genes and Chromosomes*, focusing on scientific communication and outreach.

Education

Astronomy Institute & Physics Institute — Mexico City, CDMX
M.S. in Field and Particles Astrophysics — *Aug 2024 - Current*

- **Concentrations**: Neural Networks,Cosmology & Quantum Field Theory (QFT)
- **Coursework**: Neural Networks, Astrophysics, Galactic Dynamics, Stellar Evolution, Interstellar Medium, Cosmology & QFT

Carnegie Mellon University — Online Course
Introduction to Modern AI — *Jan 2026 - April 2026*

- **Concentrations**: Large language models and Neural Networks
- **Relevant Coursework**: Supervised Machine Learning, Reinforcement learning, AI safety and security.

National Autonomous Mexico University — Mexico City, CDMX
B.S. in Physics, Graduated with honours for high academic performance. — *Aug 2019 - Dec 2023*

- **Concentrations**: Theoretical physics, Data Science & Machine Learning
- **Relevant Coursework**: Advanced Calculus, Linear Algebra, Statistics, Data Science, Theoretical physics, Fluids simulations, Advanced Mathematical Methods, AI modelling.

National Autonomous Mexico University — Mexico City, CDMX
Finance Coursework — *March 2026 - Current*

- **Concentrations**: Business finances, investments, credits and stock market
- **Relevant Coursework**: Financial mathematics, stock market technical analysis (moving averages, MACD), administration and accountability basics